Syracuse University
School of Information Studies

Week 6
Introduction to Visualization

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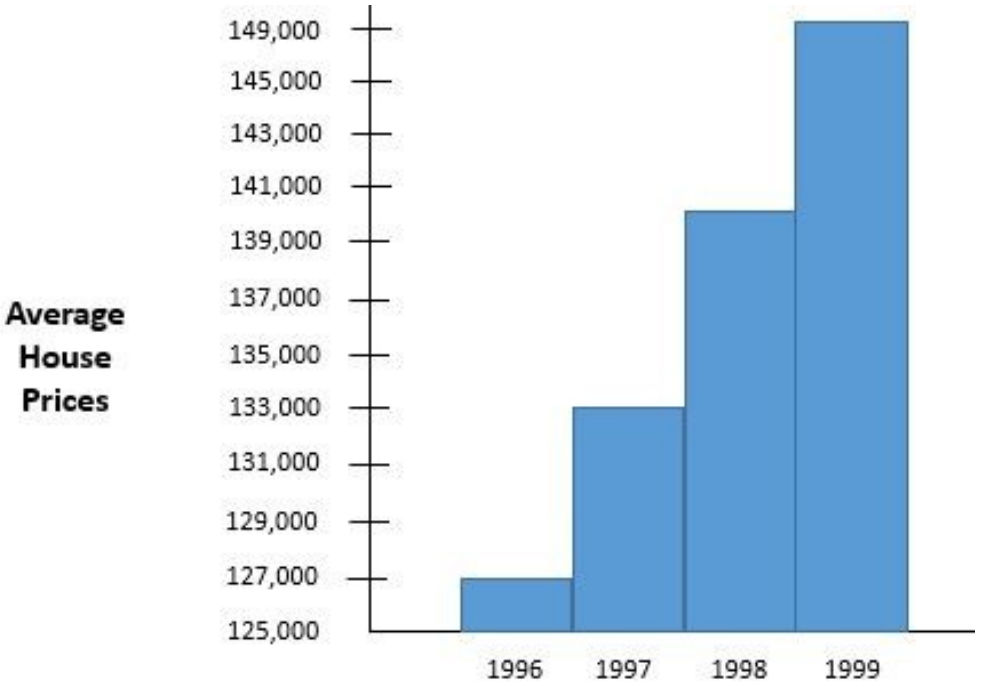
Definitions

- **Information visualization** can be defined as, “The creation and use of interactive visual representations of abstract data.”

- **Analyzing Visualizations:**

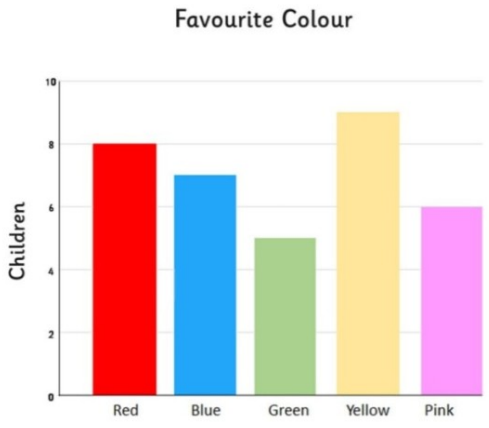


Bar Chart

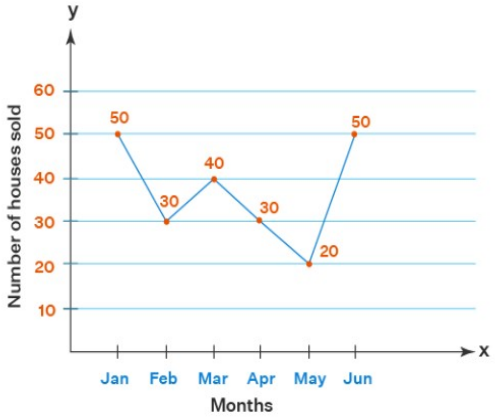
The Three Important Charts

Data scientists believe you can generally answer 80% of all data questions by using these three visualization chart types.

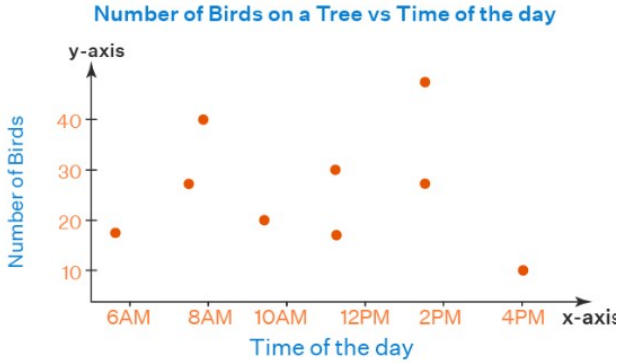
- **Bar Chart:** Used to compare categories.



- **Line Chart**
  - Used to show change over time.
  - **Example:** Temperature over days



- **Scatter Plot**
  - Used to see relationships between two numbers.
  - **Example:** Study hours vs exam score



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The ggplot2 package

ggplot2

ggplot2 divides plot into three different fundamental parts:

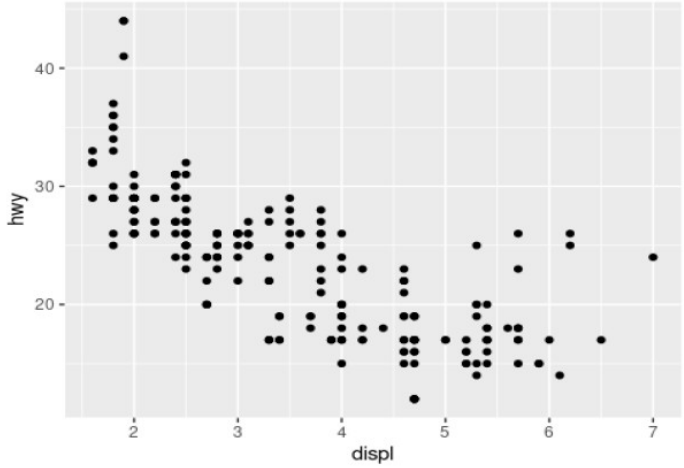
Plot = Data + Aesthetics + Geometry

Every plot can be defined as follows:

- **Data** is a data frame.
- **Aesthetics** is used to indicate x and y variables. It can also be used to control the color, the size or the shape of points, the height of bars, etc.
- **Geometry** defines the type of display (histogram, box plot, line plot, density plot, dot plot, etc.)

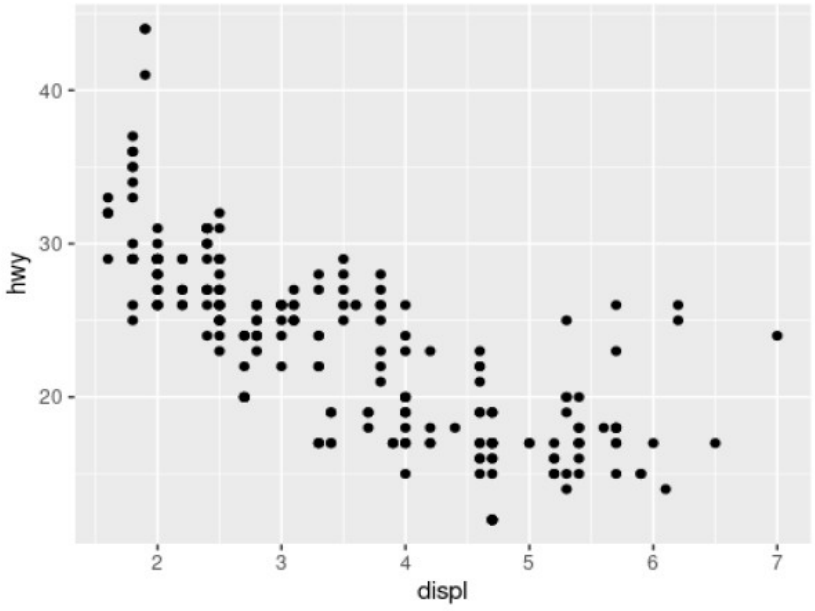
Working with GGPlot2:

```
install.packages("ggplot2")
library(ggplot2)

View(mpg)

myPlot <- ggplot(mpg)

# The aesthetic
myPlot <- myPlot + aes(x=displ,y=hwy)

# The geometry
myPlot <- myPlot + geom_point()

myPlot
```



- **ggplot** is a function from the **ggplot2 library** in R.
- **mpg** is a built-in dataset about car fuel economy (miles per gallon, car type, engine size, etc.).
- **myPlot <- ggplot(mpg)**
  - Create a new plot object using the **mpg** dataset and save it in **myPlot.**
- The aesthetic: **myPlot <- myPlot + aes(x=displ,y=hwy)**
  - **aes(...)** means aesthetics. It tells **ggplot** which columns of the dataset go on the **x-axis** and **y-axis.**
  - **x = displ** : Put **engine size (displacement)** on the x-axis.
  - **y = hwy** : Put **highway miles per gallon** on the y-axis.
- The geometry: **myPlot <- myPlot + geom_point()**
  - **geom_point ()** means draw the data as points. This makes a scatter plot.
  - **Each dot in the plot = one car from the mpg dataset**.

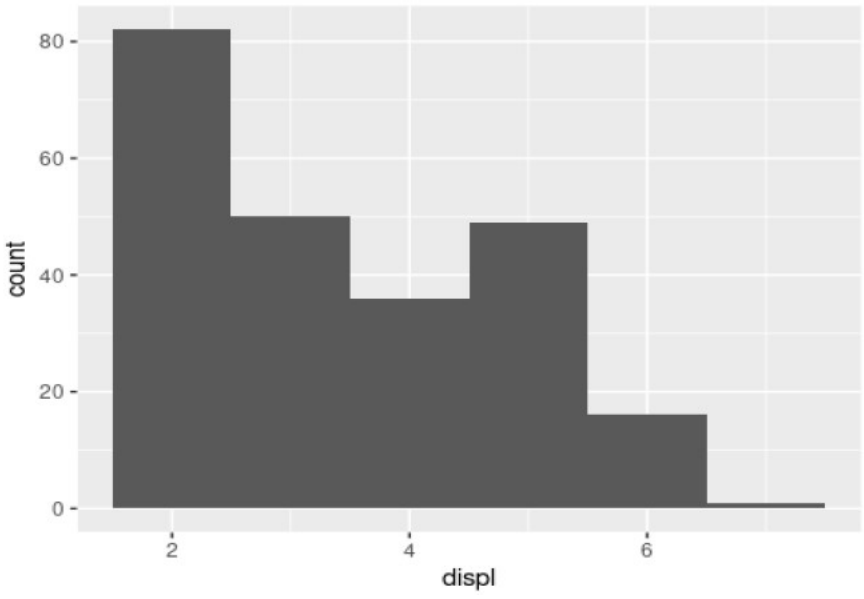
ggplot(mpg) + aes(x = displ, y = hwy) + geom_point()



- The above line is the short version of what you built step by step before.
- **ggplot(mpg):** Use the **mpg dataset.**
- **aes(x = displ, y = hwy)** : Map **engine size to the x-axis** and **highway mpg to the y-axis.**
- **geom_point ()** : Draw the data as points, gives you a scatter plot.

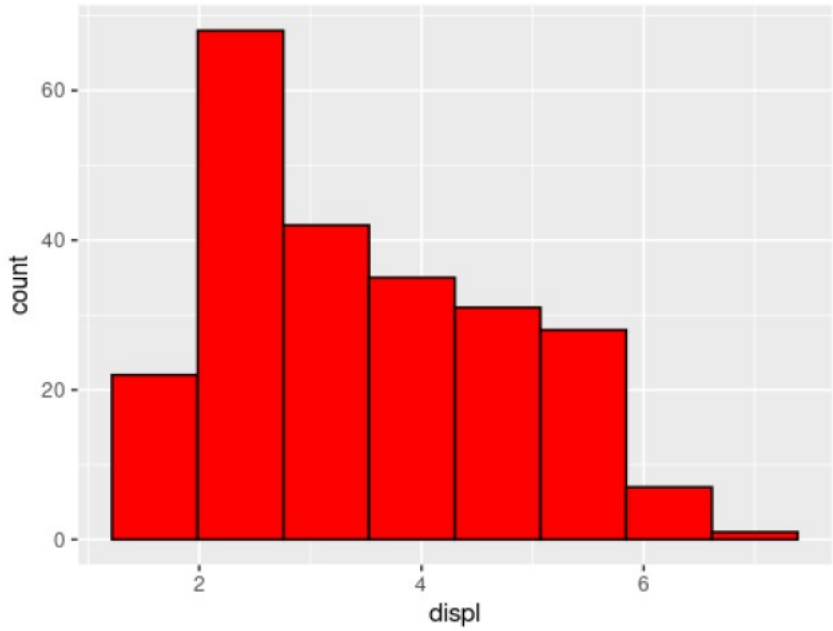
Univariate Display: Control Binwidth

```
mpg %>%
  ggplot() +
  aes(x=displ)+
  geom_histogram(binwidth=1)
```



- **mpg %>%** : the pipe (%>%) sends the **mpg dataset** into **ggplot().**
- It's just another way of writing **ggplot(mpg).**
- **ggplot() :** Start a plot with the **mpg data.**
- **aes(x = displ):** Put engine size (displacement) on the x-axis.
- **geom_histogram():** Make a histogram.
- **The result:** a bar-like plot where each bar = number of cars with engine size in that range.
- By default, **geom_histogram()** picks its own bin width. Sometimes that's too wide or too narrow. You can control it with:
- **binwidth = 1** : Each bar covers an engine size range of 1 liter. For example: 1–2 L, 2–3 L, 3–4 L, etc.
- This makes the histogram easier to read because you control how "chunky" or "smooth" the bars are.

```r
mpg %>%
  ggplot() +
  aes(x=displ)+
  geom_histogram(bins=8,
      fill="red",col="black")
```

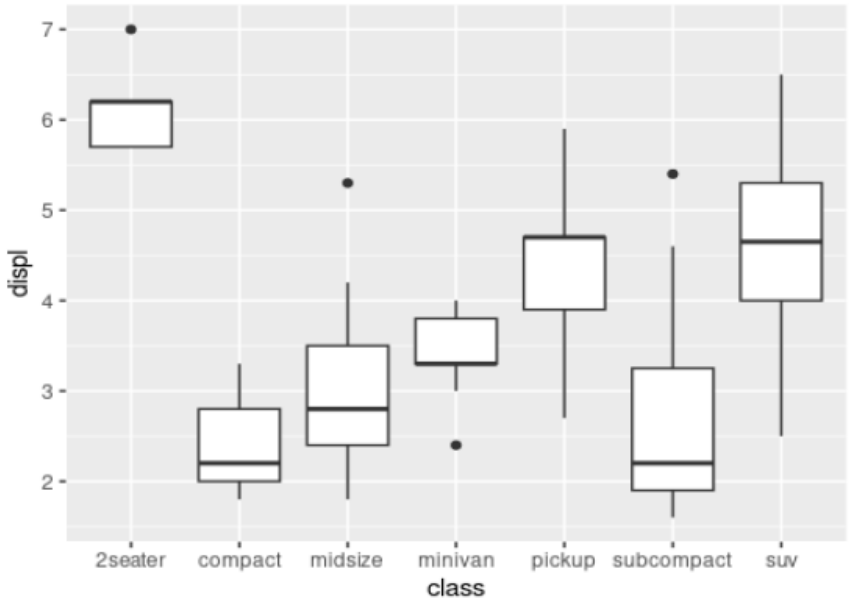


- **fill = "red"** : The inside of each bar will be red.
- **col = "black"** : The outline (border) of each bar will be black.

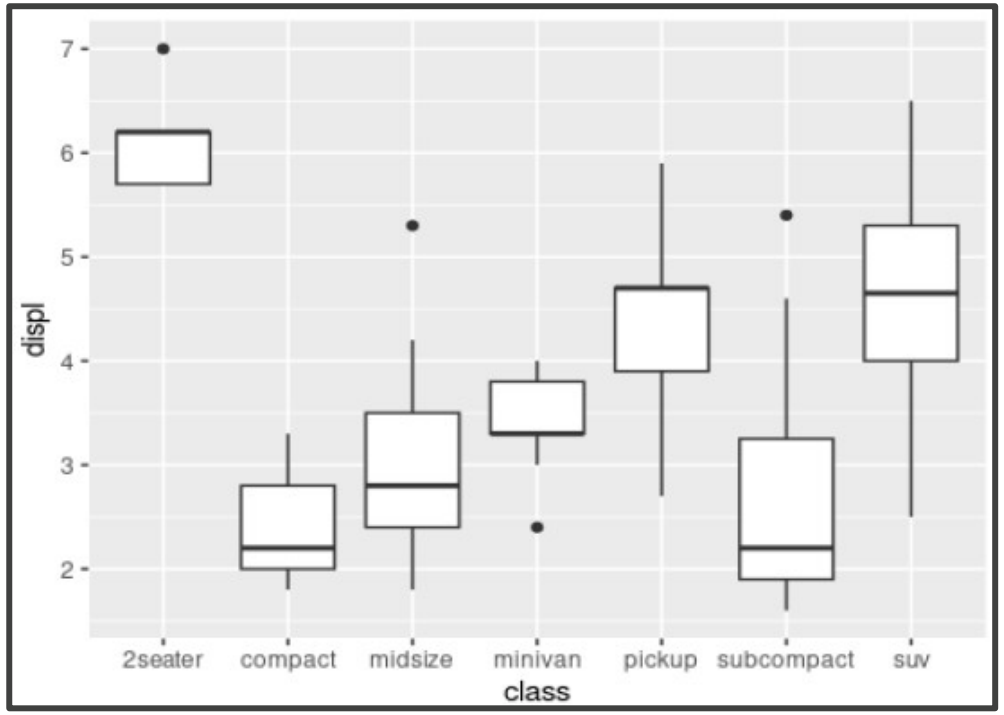
Note: ggtitle() → sets the **title**

Boxplot

```
mpg %>%
  ggplot() + aes(x=class,y=displ) +
    geom_boxplot()
```



- This code creates a boxplot that shows how big the engine is for different types of cars.
- **aes(x=class, y=displ):** It puts **the type of car** (like SUV, compact, etc. class) along the **X-axis.** It puts the **engine size** in liters (displ for displacement) up the **Y-axis.**
- **geom_boxplot():** It draws the actual boxplots.



This boxplot is showing **Engine size (displ)** for different **car types (class)**

How to read ONE box?

- Let's look at just one, say **SUV**.
- Each box has 5 important parts:

1. The line inside the box

- That line is the **median** (middle value).

2. The bottom of the box

25% of SUVs have engine size below this.

3. The top of the box

75% of SUVs have engine size below this.

4. The vertical lines (whiskers)

They show the smallest and largest values (except outliers).

5. Dots

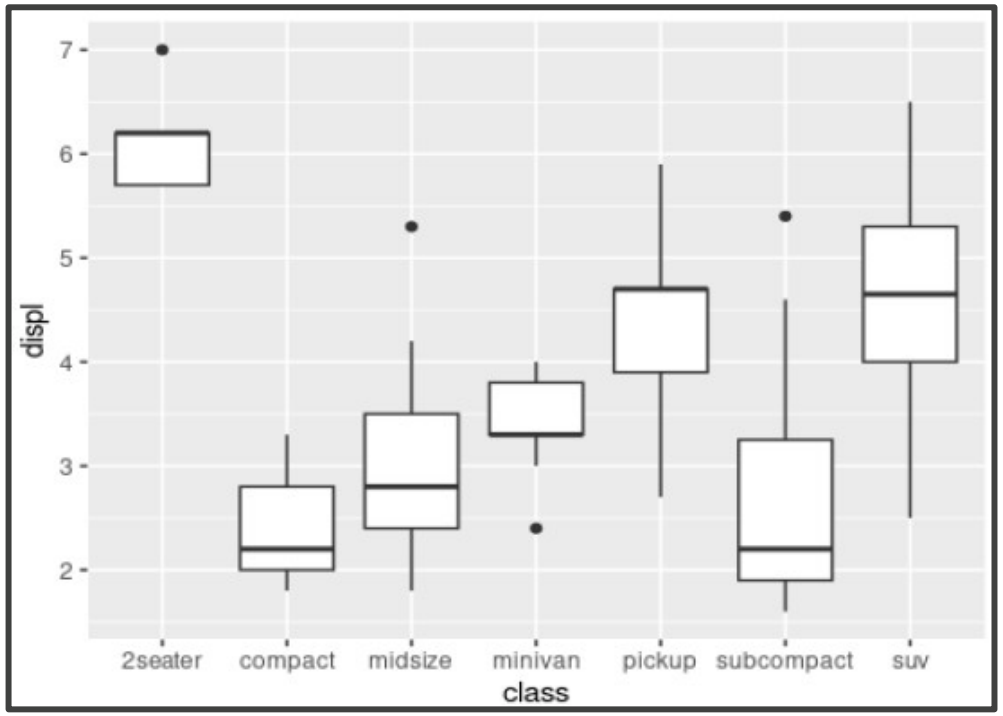
Those are **outliers** (unusual values).

What if the median and the top of the box overlap?

- ❑ That means: They are extremely close.
- ❑ Many values in the upper half are the same or very similar.

Simple Example

- Imagine engine sizes are: 3.8, 4.0, 4.2, 4.7, 4.7, 4.7, 4.7
- The upper half has many repeated 4.7 values.
- So: Median is 4.7, top of the box is also 4.7
- The median line touches the top of the box.



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The End